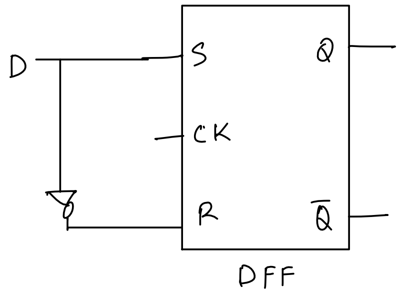
- Design a D FF using SR FF.
- Design a T FF using JK FF.
- Design a D FF using JK FF.
- Design a SR FF.

1



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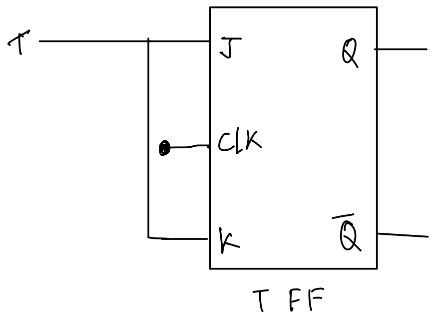

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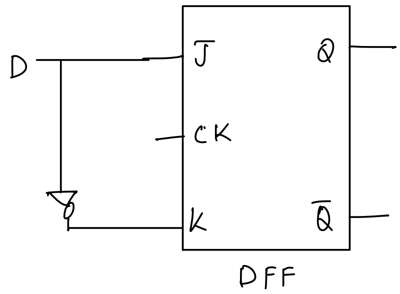

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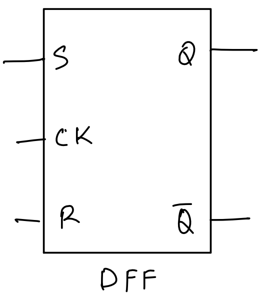
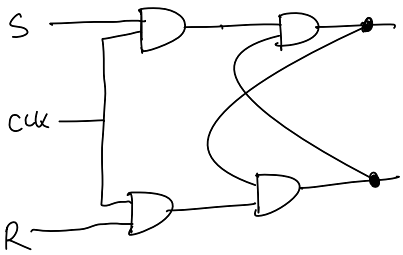
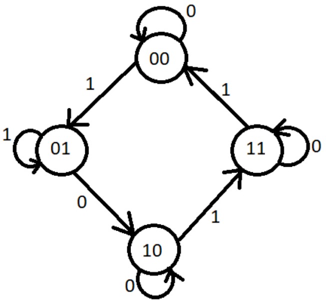




Given the state diagram as follows, get the sequential circuit using SR flipflop.



Ans:

A	B	x	A^+	B^+	S_A	R_A	S_B	R_B	
0	0	0	0	0	0	X	0	X	0
0	0	1	0	1	0	X	1	0	1
0	1	0	1	0	1	0	0	1	2
0	1	1	0	1	0	X	X	0	3
1	0	0	1	0	X	0	0	X	4
1	0	1	1	1	X	0	1	0	5
1	1	0	1	1	X	0	X	0	6
1	1	1	0	0	0	1	0	1	7

For S_A,

$A \backslash Bx$	$B'x'$	$B'x$	Bx	Bx'
A'	0	1	3	1 2
A	X 4	X 5	7	X 6

$S_A = Bx'$

For S_B,

$A \backslash Bx$	$B'x'$	$B'x$	Bx	Bx'
A'	0	1 1	X 3	2
A	4	1 5	7	X 6

$S_B = B'x$

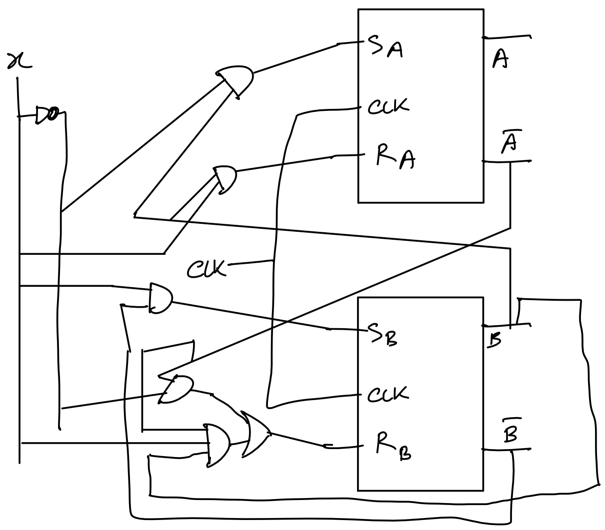
For R_A,

$A \backslash Bx$	$B'x'$	$B'x$	Bx	Bx'
A'	X 0	X 1	X 3	2
A	4	5	1 7	6

$R_A = Bx$

For R_B,

$A \backslash Bx$	$B'x'$	$B'x$	Bx	Bx'
A'	X 0	1	3	1 2
A	X 4	5	1 7	6



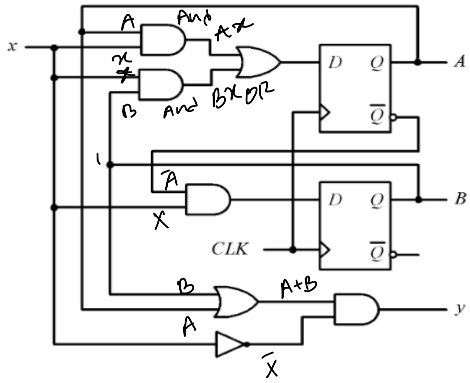
$R_B = A'x' + ABx$

$R_A = Bx$

$S_B = B'x$

$S_A = Bx'$

- Draw the state diagram for the given circuit.

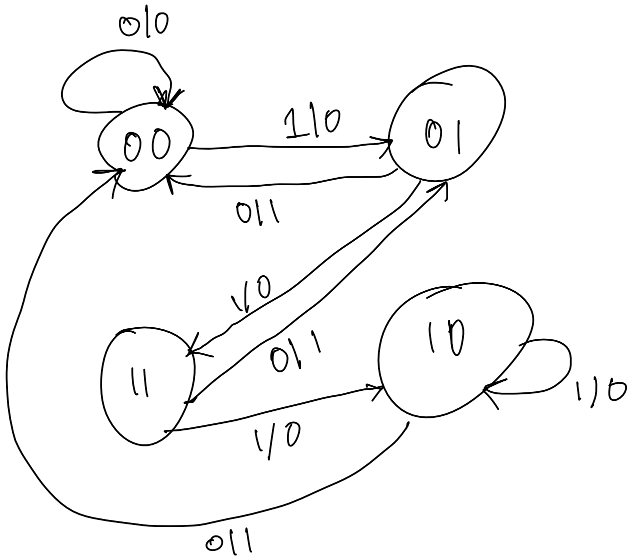


$D_A = Ax + Bx$

$D_B = \bar{A}x$

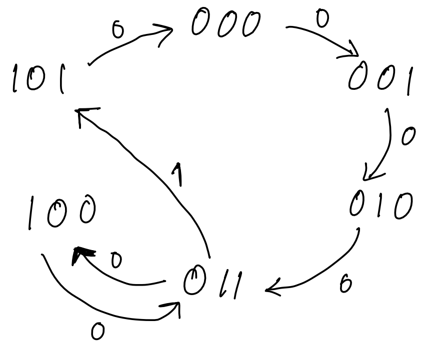
$y = (A+B) \cdot \bar{x}$

A	B	x	D_A	D_B	y	A^+	B^+
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1
0	1	0	0	0	1	0	0
0	1	1	1	1	0	1	1
1	0	0	0	0	1	0	0
1	0	1	1	0	0	1	0
1	1	0	0	0	1	0	1
1	1	1	1	0	0	1	0



CSE110 → CSE111 → CSE220 → CSE221 → CSE331 → CSE221 → CSE321 → CSE110
1 2 3 4 5 4 6 1

CSE110 → 0 0 0
CSE111 → 0 0 1
CSE220 → 0 1 0
CSE221 → 0 1 1
CSE331 → 1 0 0
CSE321 → 1 0 1



0	0	0
0	1	1
1	0	1
1	1	0

x	A	B	C	A^+	B^+	C^+	T_A	T_B	T_C	
0	0	0	0	0	0	1	0	0	1	0
0	0	0	1	0	1	0	0	1	1	1
0	0	1	0	0	1	1	0	0	1	2
0	0	1	1	1	0	0	1	1	1	3
0	1	0	0	0	1	1	1	1	1	4
0	1	0	1	0	0	0	1	0	1	5
0	1	1	0	X	X	X	X	X	X	6
0	1	1	1	X	X	X	X	X	X	7
1	0	0	0	0	0	1	0	0	1	8
1	0	0	1	0	1	0	0	1	1	9
1	0	1	0	0	1	1	0	0	1	10
1	0	1	1	1	0	1	1	1	0	11
1	1	0	0	0	1	1	1	1	1	12
1	1	0	1	0	0	0	1	0	1	13
1	1	1	0	X	X	X	X	X	X	14
1	1	1	1	X	X	X	X	X	X	15

For T_A,

$xA \backslash BC$	$B'C'$	$B'C$	BC	BC'
$x'A'$	0	1	1 3	2
$x'A$	1 4	1 5	X 7	X 6
xA	1 12	1 13	X 15	X 14
xA'	8	9	1 11	10

$$T_A = A + BC$$

For T_B,

$xA \backslash BC$	$B'C'$	$B'C$	BC	BC'
$x'A'$	0	1 1	1 3	2
$x'A$	1 4	5	X 7	X 6
xA	1 12	13	X 15	X 14
xA'	8	1 9	1 11	10

$$T_B = AC' + A'C$$

$$\Rightarrow T_B = A \oplus C$$

For T_C,

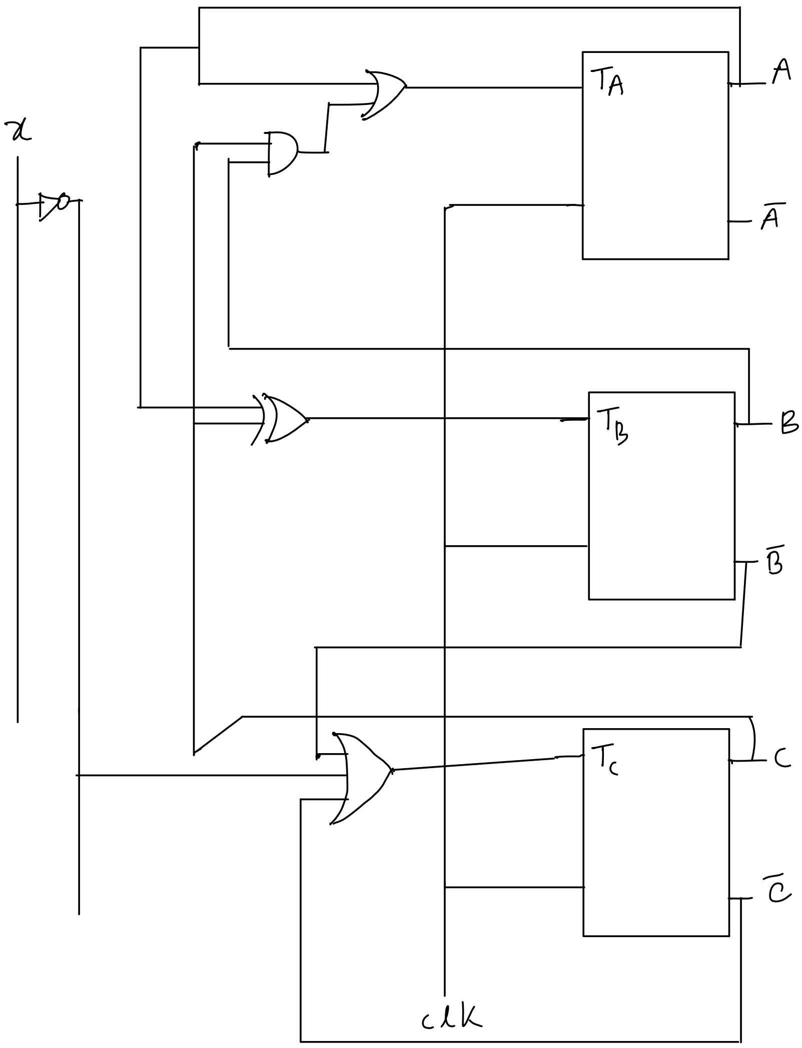
$xA \backslash BC$	$B'C'$	$B'C$	BC	BC'
$x'A'$	1 0	1 1	1 3	1 2
$x'A$	1 4	1 5	X 7	X 6
xA	1 12	1 13	X 15	X 14
xA'	1 8	1 9	11	1 10

$$T_C = B' + x' + C'$$

$T_C = B' + x' + C'$

$T_A = A + BC$

$\Rightarrow T_B = A \oplus C$

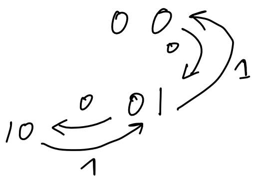


- Implement the following counter using JK FF: Green->Yellow->Red->Yellow->Green

Green → 00

Yellow → 01

Red → 10



For J_A,

$x \backslash AB$	$A'B'$	$A'B$	AB	AB'
x'	0	1 1	X 3	X 2
x	4	5	X 7	X 6

$J_A = x'B$

	x	A	B	A^+	B^+	J_A	K_A	J_B	K_B
0	0	0	0	0	1	0	X	1	X
1	0	0	1	1	0	1	X	X	1
2	0	1	0	0	1	X	1	1	X
3	0	1	1	X	X	X	X	X	X
4	1	0	0	0	1	0	X	1	X
5	1	0	1	0	0	0	X	X	1
6	1	1	0	0	1	X	1	1	X
7	1	1	1	X	X	X	X	X	X

For K_A,

$x \backslash AB$	$A'B'$	$A'B$	AB	AB'
x'	X 0	X 1	X 3	1 2
x	X 4	X 5	X 7	1 6

$K_A = 1$

For J_B,

$x \backslash AB$	$A'B'$	$A'B$	AB	AB'
x'	1 0	X 1	X 3	1 2
x	1 4	X 5	X 7	1 6

$J_B = 1$

For K_B,

$x \backslash AB$	$A'B'$	$A'B$	AB	AB'
x'	X 0	1 1	X 3	X 2
x	X 4	1 5	X 7	X 6

$K_B = 1$